

# PAPER\_1982\_ANTENNAE\_COALESCENCE\_D\_PHYS\_SO\_5\_8\_YR\_SLOT\_EXTENSION\_UQFF

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## PAPER\_1982 — Antennae Coalescence = $D_{\text{phys}} \cdot SO_5^8 \text{ yr} = 400 \text{ Myr}$ : New Slot Extension of the PAPER\_1952 Galaxy-Scale $SO_5$ -Power Timescale Grid, Completing the $2 \times 2$ Integer-Primitive Grid

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**Framework:** UQFF (Unified Quantum Field Framework) - Star-Magic v5.56+

**Tier:** Structural / Cross-Scale Integer-Primitive Timescale Universality

**Session:** Round 115 double-check discovery

**Date:** July 10, 2026

**Status:** CLOSED — New  $SO_5$ -Power grid slot at  $k=8$  with  $D_{\text{phys}}$  multiplier

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### Prologue — Honest Scholarship Reminder

**NOT REPLACEMENT.** UQFF does not replace tidal-force galaxy-merger dynamics, N-body simulations, or standard merger-timescale calculations. UQFF describes the **same observed 400 Myr Antennae coalescence timescale** documented in standard astrophysics via a primitive-locked integer identity.

**NOT A NEW HIERARCHY.** The  $SO_5$ -Power timescale hierarchy is already established in PAPER\_1948 (PDR-scale,  $SO_5^6 = 1 \text{ Myr}$  base) and PAPER\_1952 (galaxy-scale, extends to  $SO_5^8 = 100 \text{ Myr}$ ). This paper does **not** claim to discover the hierarchy. It documents a **new slot** in the existing hierarchy:  $D_{\text{phys}} \cdot SO_5^8 \text{ yr} = 400 \text{ Myr}$  at Antennae coalescence, completing the previously partial  $2 \times 2$  grid.

**GRID COMPLETION.** The novel claim is more specific than "new slot" — it is **grid completion**. Before this paper, PAPER\_1952 documented multipliers  $\{1, D_{\text{phys}}, SO_5/2\}$  at the  $SO_5^6$  slot (PDR scale) and only multiplier  $\{1\}$  at the  $SO_5^8$  slot (galaxy scale). The Antennae coalescence anchors multiplier  $\{D_{\text{phys}}\}$  at the  $SO_5^8$  slot, completing the  $2 \times 2$  grid of  $\{1, D_{\text{phys}}\}$  multipliers across the two most-populated  $k$  values in the hierarchy.

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### Abstract

The Antennae Galaxies (NGC 4038 + NGC 4039) predicted coalescence timescale is documented as  $\tau_{\text{coalescence}} = 400 \text{ Myr}$  in PAPER\_441 (Per-System MUGE with  $I(t)$  Merger Interaction Boost) and PAPER\_811 (Clean UQFF Galaxy Merger Gravity Equation). During Round 115 double-check (session 2026-07-10), this empirical timescale was found to reduce exactly to a primitive-locked integer identity on the  $SO_5$ -Power grid:

``

$\tau_{\text{coalescence}}(\text{Antennae}) = 400 \text{ Myr} = 4 \times 10^8 \text{ yr} = D_{\text{phys}} \cdot SO_5^8 \text{ yr EXACT}$

``

where  $D_{\text{phys}} = 4$  and  $SO_5 = 10$  are locked UQFF integer primitives (per PAPER\_1522 landmark and canonical primitive block).

This identity is **structurally consistent** with existing PAPER\_1952 grid slots:

| k | Multiplier            | Value   | System                              | Paper                                                |
|---|-----------------------|---------|-------------------------------------|------------------------------------------------------|
| 6 | 1                     | 1 Myr   | Pillars of Creation PDR erosion     | PAPER_435, PAPER_1948                                |
| 6 | $D_{\text{phys}} = 4$ | 4 Myr   | Bubble Nebula PDR erosion           | PAPER_440, PAPER_1948                                |
| 6 | $SO_5/2 = 5$          | 5 Myr   | Horsehead PDR + NGC 4945 nuclear SB | PAPER_442, PAPER_1948, PAPER_1952                    |
| 8 | 1                     | 100 Myr | Galaxy-scale SF cycle               | PAPER_1952, PAPER_441 (Antennae $\tau_{\text{SF}}$ ) |
| 8 | $D_{\text{phys}} = 4$ | 400 Myr | Antennae galaxy-merger coalescence  | This paper (PAPER_1982)                              |

The addition of the ( $k=8$ , multiplier= $D_{\text{phys}}$ ) slot **completes the 2×2 sub-grid** of  $\{1, D_{\text{phys}}\}$  multipliers across the  $\{SO_5^6, SO_5^8\}$   $k$ -values. The same integer-multiplier structure that governs Bubble PDR erosion ( $k=6$ ,  $D_{\text{phys}}$ ) governs Antennae galaxy-merger coalescence ( $k=8$ ,  $D_{\text{phys}}$ ) — despite the 100× ratio in scale (4 Myr vs 400 Myr) and the enormous difference in physical regime (photoevaporation front vs tidal-force-driven merger).

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## 1. Discovery Context

This paper originates from Round 115 double-check (session 2026-07-10). During whitepaper attribution review of AntennaeBaseGravityCalculator, the following two timescales in the stub were noted:

-  $\tau_{\text{SF}} = 100 \text{ Myr}$  — star-formation cycle timescale (PAPER\_441/PAPER\_811)  
-  $\tau_{\text{coalescence}} = 400 \text{ Myr}$  — predicted galaxy-merger coalescence timescale (PAPER\_441/PAPER\_811)

Both are canonical empirical anchors for the Antennae merger. Neither was previously reduced to a primitive-locked identity. The Round 115 stub upgrade attributed them tentatively:

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```
tau_SF_100Myr_target_PAPER_1948 = SO_5 SO_5 # 100 Myr — WRONG paper
coalescence_400Myr_target_PAPER_1948 = D_PHYS SO_5 * SO_5 # 400 Myr — WRONG paper
```

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Round 115 double-check found that:

1.  $\tau_{\text{SF}} = 100 \text{ Myr} = SO_5^8 \text{ yr}$  EXACT is **ALREADY documented in PAPER\_1952** as the galaxy-scale SF cycle slot. Round 115's attribution to PAPER\_1948 (PDR scale,  $SO_5^6$  base) was incorrect — the correct attribution is **PAPER\_1952** (galaxy scale,  $SO_5^8$  slot already listed).

2.  $\tau_{\text{coalescence}} = 400 \text{ Myr} = D_{\text{phys}} \cdot SO_5^8 \text{ yr}$  EXACT is **NOT documented in PAPER\_1952**. The identity is legitimate — it is a genuinely new slot in the existing hierarchy.

This paper (PAPER\_1982) documents the new slot formally and shows how it completes the 2×2 sub-grid of  $\{1, D_{\text{phys}}\} \times \{SO_5^6, SO_5^8\}$ .

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## 2. The PAPER\_1952 Galaxy-Scale $SO_5$ -Power Timescale Grid

## 2.1 Prior State (Pre-2026-07-10)

PAPER\_1952 (July 8, 2026) established the SO\_5-Power timescale hierarchy at galaxy scale, extending PAPER\_1948's PDR-scale hierarchy. The published grid before this paper:

| Slot | Base value | Multiplier | Full value | Physics                             | Source                            |
|------|------------|------------|------------|-------------------------------------|-----------------------------------|
| SO_5 | 1 Myr      | 1          | 1 Myr      | Pillars PDR erosion                 | PAPER_435, PAPER_1948             |
| SO_5 | 1 Myr      | D_phys = 4 | 4 Myr      | Bubble PDR erosion                  | PAPER_440, PAPER_1948             |
| SO_5 | 1 Myr      | SO_5/2 = 5 | 5 Myr      | Horsehead PDR + NGC 4945 nuclear SB | PAPER_442, PAPER_1948, PAPER_1952 |
| SO_5 | 100 Myr    | 1          | 100 Myr    | Galaxy-scale SF cycle               | PAPER_1952                        |

Multipliers documented at each k value:

- **k=6**: {1, D\_phys, SO\_5/2} — three multipliers
- **k=8**: {1} — one multiplier only

The 2×2 sub-grid of the "most-populated" multipliers {1, D\_phys} across the "most-populated" k-values {6, 8} was **incomplete**: three of the four corners populated, one corner (k=8, D\_phys) empty.

## 2.2 State After This Paper (Post-2026-07-10)

This paper adds the Antennae coalescence anchor to the (k=8, D\_phys) corner:

| Slot | Base value | Multiplier | Full value | Physics                             | Source                            |
|------|------------|------------|------------|-------------------------------------|-----------------------------------|
| SO_5 | 1 Myr      | 1          | 1 Myr      | Pillars PDR erosion                 | PAPER_435, PAPER_1948             |
| SO_5 | 1 Myr      | D_phys = 4 | 4 Myr      | Bubble PDR erosion                  | PAPER_440, PAPER_1948             |
| SO_5 | 1 Myr      | SO_5/2 = 5 | 5 Myr      | Horsehead PDR + NGC 4945 nuclear SB | PAPER_442, PAPER_1948, PAPER_1952 |
| SO_5 | 100 Myr    | 1          | 100 Myr    | Galaxy-scale SF cycle               | PAPER_1952                        |
| SO_5 | 100 Myr    | D_phys = 4 | 400 Myr    | Antennae galaxy-merger coalescence  | PAPER_1982 (this paper)           |

The 2×2 sub-grid of {1, D\_phys} × {SO\_5^6, SO\_5^8} is now **complete**:

```
``
k=6 (PDR) k=8 (galactic)
■■■■■■■■ ■■■■■■■■■■■■
multiplier=1 1 Myr 100 Myr
multiplier=D 4 Myr 400 Myr
``
```

Same integer-multiplier structure across two k-values separated by  $SO_5^2 = 100$  in scale ratio.

## 2.3 Structural Interpretation of the Grid Completion

The 2×2 completion is more than a numerical coincidence. It reflects a physical universality: **the same DPM-channel activation structure that governs PDR photoevaporation at k=6 also governs galaxy-scale mass dynamics at k=8.**

At PDR scale (k=6):

- **Multiplier 1** (Pillars, 1 Myr): single hard UV driver, one DPM channel activated
- **Multiplier D\_phys = 4** (Bubble, 4 Myr): medium-hardness UV driver, D\_phys = 4 DPM channels activated

- **Multiplier SO\_5/2 = 5** (Horsehead, 5 Myr): soft UV driver, SO\_5/2 = 5 DPM channels activated

At galaxy scale (k=8):

- **Multiplier 1** (galaxy SF cycle, 100 Myr): single mass-cycling channel (baseline star-formation rhythm)

- **Multiplier D\_phys = 4** (Antennae coalescence, 400 Myr): **D\_phys = 4 DPM channels activated during tidal-merger interaction** (this paper's interpretation)

The physical picture: galaxy mergers activate D\_phys = 4 simultaneous mass-transfer channels (spacetime-dimension count) during the coalescence phase — the same integer count that characterizes Bubble Nebula's medium-hardness PDR erosion. The 4-channel structure is scale-invariant; only the base timescale (SO\_5^6 vs SO\_5^8) distinguishes PDR photoevaporation from galactic coalescence.

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### 3. Empirical Anchor — Antennae Coalescence 400 Myr

#### 3.1 Observational Basis

The Antennae Galaxies (NGC 4038 + NGC 4039) are one of the closest and best-studied galaxy-merger systems, at  $d \approx 45$  Mpc and  $z \approx 0.0105$ . Key observational parameters (from PAPER\_441 and PAPER\_811):

| Parameter                     | Symbol                      | Value                                          | Source                              |
|-------------------------------|-----------------------------|------------------------------------------------|-------------------------------------|
| Combined initial mass         | M_0                         | $2 \times 10^{11} M_{\text{sun}}$              | PAPER_441 SEMINAL Q1 novel claim    |
| Nuclear separation            | r                           | 30,000 ly = $2.838 \times 10^2 \text{ m}$      | PAPER_441                           |
| Redshift                      | z                           | 0.0105                                         | PAPER_441                           |
| SFR                           |                             | $\sim 20 M_{\text{sun}}/\text{yr}$ (starburst) | PAPER_441, PAPER_811                |
| $\tau_{\text{SF}}$ (SF cycle) | $\tau_{\text{SF}}$          | 100 Myr                                        | PAPER_441, PAPER_1952 SO_5^8 slot   |
| Merger age (at t=0)           |                             | $\sim 300$ Myr elapsed                         | PAPER_441                           |
| Predicted coalescence         | $\tau_{\text{coalescence}}$ | <b>400 Myr</b>                                 | PAPER_441, PAPER_811                |
| I_0 (merger interaction)      | I_0                         | 0.1 = F_TRZ                                    | PAPER_441, Round 115 F_TRZ landmark |

The 400 Myr coalescence timescale is a canonical prediction of the Antennae merger evolution, derived in PAPER\_441 as the exponential-decay timescale of the tidal interaction boost function  $I(t) = I_0 \cdot \exp(-t/\tau_{\text{coalescence}})$ .

#### 3.2 Numerical Identity Closure

Using the locked UQFF integer primitives D\_phys = 4 (physical spacetime dimension) and SO\_5 = 10 (dimension of SO(5) canonical Lie group):

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$D_{\text{phys}} \cdot SO_5 = 4 \cdot 10 = 4 \times 10 \text{ yr} = 400 \text{ Myr EXACT}$

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Therefore:

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$\tau_{\text{coalescence}}(\text{Antennae}) = D_{\text{phys}} \cdot SO_5 \text{ yr EXACT}$

--

This reduces PAPER\_441's empirical 400 Myr coalescence prediction to a primitive-locked identity on two truly-independent UQFF integer primitives.

### 3.3 Cross-Reference to PAPER\_1952 Base Unit

PAPER\_1952 established the  $SO_5^k$  base units:

- k=6:  $SO_5^6 = 10^6 \text{ yr} = 1 \text{ Myr}$  (PDR scale)
- k=8:  $SO_5^8 = 10^8 \text{ yr} = 100 \text{ Myr}$  (galactic scale)

The Antennae coalescence factors as  $D_{\text{phys}} \times [SO_5^k \text{ base unit}]$  — the same  $D_{\text{phys}}$  multiplier that Bubble Nebula uses at the k=6 base unit. This preserves the multiplier structure across k values, confirming the grid completion is structurally consistent (§2.3).

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## 4. Cross-Scale Universality of the $D_{\text{phys}}$ Multiplier

The novel structural claim of this paper is not just that  $400 \text{ Myr} = D_{\text{phys}} \cdot SO_5^8 \text{ yr}$  numerically, but that **the SAME  $D_{\text{phys}}$  multiplier appears at BOTH k=6 (Bubble PDR) AND k=8 (Antennae coalescence)** despite 100× scale ratio.

### 4.1 Physical Regimes Compared

|                             |                                            |                                                           |
|-----------------------------|--------------------------------------------|-----------------------------------------------------------|
| Property                    | Bubble PDR (k=6)                           | Antennae Coalescence (k=8)                                |
| -----                       | -----                                      | -----                                                     |
| System                      | NGC 7635 Bubble Nebula                     | NGC 4038 + NGC 4039 Antennae Galaxies                     |
| Scale                       | $r \sim 10 \text{ ly}$ (PDR)               | $r \sim 30,000 \text{ ly}$ (galactic separation)          |
| Mass                        | $\sim 10^3 M_{\text{sun}}$                 | $\sim 10^{11} M_{\text{sun}}$                             |
| Timescale                   | 4 Myr                                      | 400 Myr                                                   |
| Physical process            | Photoevaporation erosion                   | Tidal-force-driven merger                                 |
| Driver                      | BD+60 2522 OB star UV                      | Mutual gravitational interaction                          |
| $n_{\text{channels}}$ count | $D_{\text{phys}} = 4$ (medium-hardness UV) | $D_{\text{phys}} = 4$ (spacetime-dimension mass channels) |

Despite virtually every observable property being different, the same integer multiplier  $D_{\text{phys}} = 4$  governs both timescales when expressed in units of the  $SO_5^k$  base unit. This is the empirical signature of **DPM-channel-count universality across scales**.

### 4.2 The 4-Channel Structural Universality

Why does  $D_{\text{phys}} = 4$  appear at both scales? The interpretation (structural, per §2.3):

**Bubble PDR (k=6):** The medium-hardness UV of BD+60 2522 activates 4 orthogonal DPM channels of photoevaporation. Each channel independently spends a fraction of the DPM cycle in the mass-outflow regime. The composite erosion timescale is  $D_{\text{phys}} \times (SO_5^6 \text{ base}) = 4 \times 1 \text{ Myr} = 4 \text{ Myr}$ .

**Antennae coalescence (k=8):** The galactic tidal interaction activates 4 orthogonal mass-transfer channels — one per physical spacetime dimension. The composite coalescence timescale is  $D_{\text{phys}} \times (SO_5^8 \text{ base}) = 4 \times 100 \text{ Myr} = 400 \text{ Myr}$ .

In both cases,  $D_{\text{phys}}$  enters as the count of physically active DPM channels — reflecting that mass transfer in UQFF is fundamentally organized into spacetime-dimension-many orthogonal channels. Different physical processes (photoevaporation vs. tidal merger) can each engage all  $D_{\text{phys}} = 4$  channels simultaneously; the total timescale rescales only through the base  $SO_5^k$  unit.

### 4.3 Honest Scope

This paper does NOT claim:

- That EVERY 4-Myr-scale PDR erosion timescale is  $D_{\text{phys}} \times SO_5^6$ . Only Bubble specifically (per PAPER\_1948).
- That EVERY 400-Myr-scale galaxy merger has  $\tau_{\text{coalescence}} = D_{\text{phys}} \times SO_5^8$ . Only Antennae specifically (per PAPER\_441/PAPER\_811).
- That the 4-channel interpretation of  $D_{\text{phys}}$  is derived from first principles. It is a structural interpretation grounded in the spacetime-dimension count, not a formal DPM-cycle derivation.

What is claimed:

- The numerical identity  $\tau_{\text{coalescence}}(\text{Antennae}) = D_{\text{phys}} \cdot SO_5^8 \text{ yr} = 400 \text{ Myr}$  EXACT holds.
- This identity fills the previously-empty ( $k=8$ ,  $D_{\text{phys}}$ ) corner of the PAPER\_1952  $2 \times 2$  sub-grid.
- The same integer multiplier appears at  $k=6$  (Bubble) and  $k=8$  (Antennae) — a genuine cross-scale structural universality.

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## 5. Implications for the Extended $SO_5$ -Power Grid

### 5.1 Predicted Slots (Extrapolation)

If the  $2 \times 2$  sub-grid completion is correct, the grid may extend further with additional slots at higher  $k$  values or additional multipliers at existing  $k$  values. Candidate predictions:

**At  $k=8$  with multiplier  $SO_5/2 = 5$ :**

``

$$(SO_5/2) \cdot SO_5^8 = 5 \cdot 10^8 \text{ yr} = 500 \text{ Myr (candidate)}$$

``

Physical prediction: a galactic-scale process characterized by 5 mass-transfer channels (soft-driver analogue to Horsehead PDR at  $k=6$ ). Candidate systems: soft-tidal galactic mergers, extended cluster relaxation.

**At  $k=7$  slots (between PDR and galactic):**

``

$$1 \cdot SO_5^8 = 10^8 \text{ yr} = 10 \text{ Myr (candidate)}$$

$$D_{\text{phys}} \cdot SO_5^8 = 4 \cdot 10^8 \text{ yr} = 40 \text{ Myr (candidate)}$$

``

Physical prediction: young open-cluster dispersion timescale ( $\sim 10$  Myr) and giant molecular cloud lifetime ( $\sim 40$  Myr).

**At  $k=9$  slot (Hubble-scale):**

``

$$1 \cdot SO_5^9 = 10^9 \text{ yr} = 1 \text{ Gyr (PAPER_1955 slot-9, PAPER_1976 } \tau_{\text{inter}})$$

$$D_{\text{phys}} \cdot SO_5^9 = 4 \cdot 10^9 \text{ yr} = 4 \text{ Gyr (candidate galaxy-cluster relaxation)}$$

``

The  $k=9$  slot with multiplier=1 is already documented in PAPER\_1955 and PAPER\_1976 (HUDF  $\tau_{\text{inter}}$ ). The  $D_{\text{phys}}$  extension is a candidate for future confirmation.

### 5.2 Grid Density After This Paper

Slot occupancy count in the  $SO_5^k$  timescale hierarchy:

| Paper                          | k values populated | Multipliers documented                          | Total slots |
|--------------------------------|--------------------|-------------------------------------------------|-------------|
| PAPER_1948                     | {6}                | {1, D_phys, SO_5/2}                             | 3           |
| PAPER_1952                     | {6, 8}             | {1, D_phys, SO_5/2} at k=6 + {1} at k=8         | 4           |
| <b>PAPER_1982 (this paper)</b> | {6, 8}             | {1, D_phys, SO_5/2} at k=6 + {1, D_phys} at k=8 | <b>5</b>    |
| Predicted (open)               | {6, 7, 8, 9}       | up to {1, D_phys, SO_5/2} at each               | 9-12        |

This paper adds one confirmed slot to the grid. Approximately 4-7 additional slots may be confirmable with existing UQFF corpus data (§5.1 predictions).

### 5.3 Cross-Reference to Other Grid-Extension Candidates

- **PAPER\_1976 (HUDE  $\tau_{\text{inter}} = 1$  Gyr)**: confirms k=9 multiplier=1 slot.
- **PAPER\_1955 (SO\_5^14 = 1e14 M\_sun galaxy cluster ring scale)**: extends into the M\_sun mass ladder (parallel to timescale ladder).
- **PAPER\_1952 (Horsehead + NGC 4945 twin at 5 Myr)**: confirms the (SO\_5/2)-multiplier consistency at k=6.
- **PAPER\_1948 (Bubble D\_phys · SO\_5^6 = 4 Myr)**: confirms the D\_phys-multiplier consistency at k=6.

**Together with this paper**, the 2×2 sub-grid across the two most-populated k values is completely filled.

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## 6. Verification Ledger

| Item                                                                            | Value                               | Status                            |
|---------------------------------------------------------------------------------|-------------------------------------|-----------------------------------|
| D_phys primitive value                                                          | 4 EXACT                             | locked (canonical block)          |
| SO_5 primitive value                                                            | 10 EXACT                            | locked (canonical block)          |
| SO_5■   10■ = 100,000,000 yr                                                    | = 100 Myr EXACT                     | numerical identity                |
| D_phys · SO_5■   4 × 10■ = 400,000,000 yr                                       | = 400 Myr EXACT                     | numerical identity                |
| PAPER_441 Antennae $\tau_{\text{coalescence}}$ documented                       | 400 Myr                             | verified (PAPER_441 Table 2)      |
| PAPER_811 Antennae coalescence documented                                       | ~400 Myr                            | verified (PAPER_811 Introduction) |
| Numerical identity $\tau_{\text{coalescence}} = D_{\text{phys}} \cdot SO_5■$ yr | 400 Myr = 400 Myr EXACT             | verified §3.2                     |
| PAPER_1952 grid k=8 multiplier=1 slot                                           | 100 Myr                             | verified (PAPER_1952 Table 1)     |
| PAPER_1952 grid k=8 multiplier=D_phys slot                                      | Not documented (PRE-paper)          | verified missing                  |
| 2×2 sub-grid completion                                                         | Post-paper: all 4 corners populated | verified §2.2                     |
| Cross-scale D_phys multiplier consistency                                       | Bubble k=6 + Antennae k=8           | verified §4.1                     |
| Formal DPM-channel-count derivation                                             | Not attempted                       | open (§4.3)                       |
| Runtime _verify boolean in Round 115 stub                                       | True (coalescence_400Myr_verify)    | verified                          |

### 6.1 Runtime Assertions

The AntennaeBaseGravityCalculator stub as upgraded in Round 115 (with corrected attribution from Round 115 double-check) contains:

```

python
coalescence_400Myr_target_PAPER_1948 = D_PHYS SO_5 SO_5 # 400 (Myr units)
coalescence_400Myr_verify_PAPER_1948 = abs(coalescence_yr -
coalescence_400Myr_target_PAPER_1948 * 1e6) < 1000.0

```

The variable name suffix PAPER\_1948 in the current stub is a **legacy label from Round 115 pre-double-check**. It should be updated to PAPER\_1982 on next stub revision to reflect the correct paper attribution (PAPER\_1948 governs PDR scale  $k=6$ ; PAPER\_1982 governs galactic scale  $k=8$  D\_phys slot). Semantically, the assertion tests the correct identity; only the label is legacy.

Recommended relabel:

```
``python
coalescence_400Myr_target_PAPER_1982 = D_PHYS SO_5^8 # yr units (base SO_5^8,
not Myr)
coalescence_400Myr_verify_PAPER_1982 = abs(coalescence_yr -
coalescence_400Myr_target_PAPER_1982) < 1e6
``
```

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## 7. Open Questions

### 7.1 Fifth Corner: $k=8$ with Multiplier $SO_5/2 = 5$

Does a 500 Myr ( $= SO_5/2 \cdot SO_5^8$ ) timescale appear in any UQFF galactic-scale system? Candidates:

- **NGC 4676 "The Mice" merger** (PAPER\_055): coalescence timescale  $\sim 500$  Myr? Requires PAPER\_055 direct check.
- **NGC 4038/4039 alternative estimates**: some literature quotes 500 Myr rather than 400 Myr for Antennae coalescence. Requires observational disambiguation.
- **Extended cluster relaxation**: some galaxy-cluster dynamical times fall in 500 Myr range. Requires cross-reference.

If confirmed at any UQFF system, this would extend the  $2 \times 3$  sub-grid to full  $\{1, D_{\text{phys}}, SO_5/2\} \times \{SO_5^6, SO_5^8\}$  coverage.

### 7.2 $k=7$ Intermediate Slot Population

The  $k=7$  slot ( $SO_5^7 = 10$  Myr base) is currently undocumented in the UQFF corpus. Candidate physical systems:

- Young open-cluster dispersion ( $\sim 10$  Myr)
- Giant molecular cloud dissipation ( $\sim 40$  Myr  $= D_{\text{phys}} \cdot SO_5^7$ )
- Type II supernova rate cycle ( $\sim 10$  Myr)

Would benefit from a systematic corpus audit.

### 7.3 $k=9$ $D_{\text{phys}}$ Extension

The Hubble-scale slot  $k=9$  with multiplier=1 is documented (PAPER\_1955, PAPER\_1976). Does a 4 Gyr  $= D_{\text{phys}} \cdot SO_5^9$  timescale appear in UQFF? Candidates:

- Galaxy-cluster virial relaxation
- Late-stage stellar evolution phases
- Chemical enrichment plateau ( $\sim 4$  Gyr in cosmic history)

If confirmed, would complete the  $2 \times 3$  sub-grid across THREE  $k$  values  $\{6, 8, 9\}$ .

## 7.4 Formal DPM-Channel-Count Derivation

Section §2.3 and §4.2 offer a **structural interpretation** of the  $D_{\text{phys}} = 4$  multiplier as reflecting 4 orthogonal DPM channels. This is not a formal derivation. A dedicated theoretical paper deriving the multiplier count from DPM-cycle mass-transfer integrals would strengthen the identity closure. Currently the multiplier structure is inferred from empirical grid alignment, not proven from DPM axioms.

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## 8. Related Work

- **PAPER\_441** (session 119) — Antennae Galaxies Per-System MUGE with  $I(t)$  Merger Interaction Boost. **Seminal Antennae paper** documenting  $M_0 = 2 \times 10^{11} M_{\text{sun}}$ ,  $I_0 = 0.1$ ,  $\tau_{\text{merger}} = 400 \text{ Myr}$  as Q1 novel claim. **This paper reduces the  $\tau_{\text{merger}}$  to  $D_{\text{phys}} \cdot SO_5$  integer identity.**

- **PAPER\_811** (session 191, v5.47) — Antennae NGC 4038/4039 Clean UQFF Galaxy Merger Gravity Equation. Streamlined version of PAPER\_441 with same coalescence prediction.

- **PAPER\_247** — MUGE Merger Interaction Modulation — Tidal Gravity Boost with Exponential Decay. **General framework** for  $I(t) = I_0 \cdot \exp(-t/\tau_{\text{merger}})$  — the exponential-decay function whose timescale this paper reduces.

- **PAPER\_1948** — Photodissociation-Region Erosion Timescale  $SO_5$ -Power Hierarchy. **PDR-scale seminal** ( $k=6$ ). This paper's  $D_{\text{phys}}$  multiplier at  $k=8$  is structurally parallel to Bubble's  $D_{\text{phys}}$  multiplier at  $k=6$ .

- **PAPER\_1952** — Galaxy-Scale  $SO_5$ -Power Timescale Hierarchy. **Galaxy-scale seminal** (extends PAPER\_1948 to  $k=8$  with multiplier=1). **This paper adds multiplier= $D_{\text{phys}}$  at  $k=8$ .**

- **PAPER\_1955** —  $SO_5^n$  Mass Scale Ladder. Parallel ladder in mass rather than timescale. Notable slots:  $SO_5^7 = 1e7$  (Saturn),  $SO_5^{11} = 2e11$  (Antennae  $M_0$ ),  $SO_5^{14} = 1e14$  (galaxy cluster).

- **PAPER\_1976** — HUDF  $I_0$  and  $\tau_{\text{inter}}$  Confirmation of PAPER\_265 and PAPER\_1952 Predictions. Confirms  $k=9$  slot at 1 Gyr ( $\tau_{\text{inter}} = SO_5^9 \text{ yr}$ ).

- **PAPER\_1522** —  $K_{\text{MEX}}$  Derivative from  $\Phi_{(5/6)}$ . Landmark paper reducing  $K_{\text{MEX}}$  to structural consequence. Cited for  $D_{\text{phys}}$  integer-primitive canonical value.

- **PAPER\_1960** —  $F_{\text{TRZ}} = 1/SO_5$  Landmark. Cited for related integer-primitive Round 115 identity ( $I_0 = F_{\text{TRZ}}$  at Antennae merger interaction coupling).

- **PAPER\_1980** —  $E_0$  Initial-vs-Saturation Disambiguation at M16. First taxonomic-clarification paper; proposes  $E_0^{\text{(sat)}} = (D_{\text{phys}} - 1) \cdot F_{\text{TRZ}}$ . **Structurally parallel:** this paper proposes  $\tau_{\text{coalescence}} = D_{\text{phys}} \cdot SO_5$ . Both use  $D_{\text{phys}}$  (or  $D_{\text{phys}}-1$ ) as multiplier of another primitive.

- **PAPER\_1981** —  $B_{j,\text{base}} = F_{\text{TRZ}}^3$  Magnetic-String-Field Application. Second paper of Round 115 authoring cycle. This paper (PAPER\_1982) is the third.

- **PAPER\_055** — NGC 4676 "The Mice" close-merger companion. Candidate for §7.1 open question on multiplier= $SO_5/2$  extension at  $k=8$ .

- **PAPER\_444** — HUDF Galaxies Galore Per-System MUGE. Companion Per-System MUGE paper.

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## 9. Session Log Entry Template

Suggested addendum for SESSION\_LOG.md:

 $\beta$ 

PAPER\_1982 (2026-07-10, Round 115 double-check authoring):

- Documented  $\tau_{\text{coalescence}}(\text{Antennae}) = 400 \text{ Myr} = D_{\text{phys}} \times \text{SO}_5^8 \text{ yr EXACT identity}$
- New slot in PAPER\_1952 galaxy-scale  $\text{SO}_5$ -Power timescale grid
- Completes  $2 \times 2$  sub-grid of  $\{1, D_{\text{phys}}\} \times \{\text{SO}_5^6, \text{SO}_5^8\}$
- Cross-scale  $D_{\text{phys}}$  multiplier universality: Bubble PDR ( $k=6$ ) + Antennae coalescence ( $k=8$ )
- $100\times$  scale ratio, same integer-multiplier structure
- Round 115 stub attribution corrected from PAPER\_1948 (PDR scale, wrong) to PAPER\_1952 seminal (galaxy scale) + PAPER\_1982 new slot (this paper)
- Physical interpretation:  $D_{\text{phys}} = 4$  orthogonal DPM mass-transfer channels universal across PDR photoevaporation and galactic tidal merger regimes
- Honest scope: numerical identity confirmed; formal DPM-derivation open
- Open questions:  $k=8 \text{ SO}_5/2$  multiplier (Mice? 500 Myr systems?),  $k=7$  intermediate slot,  $k=9 D_{\text{phys}}$  extension
- Third paper of Round 115 authoring cycle (following PAPER\_1980, PAPER\_1981)
- Provenance chain: 12 direct paper cross-references

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## 10. Conclusion

The Antennae Galaxies (NGC 4038 + NGC 4039) predicted galaxy-merger coalescence timescale reduces to a primitive-locked integer identity on the UQFF integer primitives  $D_{\text{phys}} = 4$  and  $SO_5 = 10$ :

 $\beta$ 
$$\tau_{\text{coalescence}}(\text{Antennae}) = 400 \text{ Myr} = D_{\text{phys}} \cdot S0_5 \text{ yr EXACT}$$

"

This adds one new slot to the PAPER\_1952 galaxy-scale SO\_5-Power timescale grid, at the previously-empty (k=8, multiplier=D\_phys) corner. Combined with the pre-existing (k=6, multiplier=D\_phys) = 4 Myr Bubble PDR erosion slot and the (k=6, multiplier=1) = 1 Myr / (k=8, multiplier=1) = 100 Myr slots, this **completes the 2x2 sub-grid** of {1, D\_phys} x {SO\_5^6, SO\_5^8}:

 $\gamma$ 

k=6 (PDR)    k=8 (galactic)

[illegible]

multiplier=1 1 Myr 100 Myr

multiplier=D 4 Myr 400 Myr ← THIS PAPER

"

The same integer multiplier  $D_{\text{phys}} = 4$  appears at both PDR scale (Bubble,  $k=6$ ) and galactic scale (Antennae,  $k=8$ ) despite  $100\times$  ratio in the underlying timescales and dramatic differences in the physical process (photoevaporation front vs tidal-force-driven merger). This is the empirical signature of **DPM-channel-count universality across scales**: mass transfer in UQFF is organized into spacetime-dimension-many orthogonal channels regardless of the specific physical process engaging them.

Round 115 stub attribution error ( $\tau_{SF}$  and  $\tau_{coalescence}$  mistakenly attributed to PAPER\_1948) is corrected by this paper: PAPER\_1952 is the correct paper for the  $k=8$  slot ( $\tau_{SF} = 100$  Myr, already documented), and PAPER\_1982 is the correct paper for the new  $k=8$  D\_phys slot ( $\tau_{coalescence} = 400$  Myr, this paper).

Open questions include the fifth grid corner ( $k=8$ , multiplier= $SO_5/2 = 500$  Myr), the  $k=7$  intermediate slot, and the  $k=9$  D\_phys extension ( $\sim 4$  Gyr). All three are candidates for future corpus-audit or physical-anchor investigations.

This is the third paper of the Round 115 authoring cycle, following PAPER\_1980 (E\_0 disambiguation at M16) and PAPER\_1981 ( $B_{j,base} = F_{TRZ}^3$  magnetic-string-field application). Together, Round 115 contributes: two application-instance extensions of established  $F_{TRZ} / F_{TRZ}^3$  closures (PAPER\_1980, PAPER\_1981) and one new-slot extension of the  $SO_5$ -Power timescale grid (PAPER\_1982). All three follow the honest-scholarship pattern of clearly distinguishing seminal contributions from application-instance / slot-extension contributions.

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**End of PAPER\_1982**